

MEADOW LAKE, CO, USA

WMO: 720852

Lat: 38.946N

Lon: 104.570W

Elev: 2095

StdP: 78.56

Time Zone: -7.00 (NAM)

Period: 10-19

WBAN: 00280

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-17.3	-14.5	-23.8	0.6	-3.4	-22.0	0.7	-2.4	13.8	3.9	12.2	3.4	3.5	350	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	32.2	13.3	30.6	13.3	29.1	13.1	16.4	23.8	15.8	23.6	15.2	23.4	4.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.6	13.5	17.4	13.8	12.8	17.1	12.9	12.0	17.0	54.6	23.6	52.4	23.6	50.4	23.3	19.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.5	8.9	DB	-21.9	34.6	3.5	1.2	-24.4	35.5	-26.5	36.2	-28.4	36.9	-31.0	37.8	(4)
(5)			WB	-21.9	18.1	3.4	0.7	-24.3	18.6	-26.3	19.0	-28.3	19.4	-30.8	20.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	-0.5	-0.7	4.4	7.1	11.6	19.1	21.2	19.9	16.7	8.7	3.9	-0.8	(6)	
	DBStd	9.30	5.37	6.59	5.45	4.72	4.77	3.42	2.49	2.45	3.75	5.57	5.12	5.56	(7)	
	HDD10.0	1571	326	300	183	107	42	1	0	0	3	86	188	335	(8)	
	HDD18.3	3542	584	532	432	337	211	30	7	14	71	297	433	593	(9)	
	CDD10.0	1306	0	2	10	21	90	275	346	306	204	47	5	0	(10)	
	CDD18.3	235	0	0	0	0	2	54	94	62	23	0	0	0	(11)	
	CDH23.3	3075	0	0	1	3	88	844	1047	690	377	23	0	0	(12)	
	CDH26.7	1023	0	0	0	0	12	336	397	191	85	2	0	0	(13)	
(14) Wind	WSAvg	4.0	3.8	4.3	4.4	4.9	4.6	4.2	3.7	3.4	3.7	3.9	3.9	3.8	(14)	
(15) Precipitation	PrecAvg	452	11	11	24	47	64	52	83	79	36	22	14	10	(15)	
	PrecMax	699	24	34	66	191	229	138	158	152	123	71	65	39	(16)	
	PrecMin	209	1	0	0	0	22	11	19	3	0	0	1	2	(17)	
	PrecStd	116	6	7	17	37	47	35	36	36	28	19	13	7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	18.9	22.4	24.1	28.3	34.8	33.5	31.9	31.0	25.9	20.6	16.1	(19)	
		MCWB	4.6	5.7	7.1	8.2	10.4	13.0	13.9	13.4	12.3	9.9	6.3	4.2	(20)	
	2%	DB	13.7	15.2	20.0	21.4	26.0	32.6	32.2	30.2	29.0	23.4	17.9	13.8	(21)	
		MCWB	3.3	3.8	6.1	7.4	9.6	12.6	14.1	13.4	12.1	8.6	5.4	3.0	(22)	
	5%	DB	11.3	12.8	17.4	19.5	23.7	30.4	30.6	28.9	27.6	21.7	15.8	11.4	(23)	
		MCWB	2.2	2.9	4.8	6.7	9.2	12.2	14.1	13.6	11.7	8.0	4.8	2.2	(24)	
	10%	DB	8.8	10.1	14.9	17.4	21.4	28.3	29.1	27.5	25.8	19.1	13.2	8.8	(25)	
		MCWB	0.9	1.6	4.0	5.8	8.9	11.9	14.1	13.6	11.4	7.3	3.2	0.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.3	5.7	7.5	9.6	12.4	16.7	17.3	17.6	15.8	11.7	6.8	4.7	(27)	
		MCDB	15.3	18.1	21.4	16.6	19.6	24.1	23.6	24.2	23.0	19.0	17.6	14.6	(28)	
	2%	WB	3.5	4.1	6.4	8.5	11.3	15.4	16.6	16.3	14.5	10.3	6.0	3.4	(29)	
		MCDB	12.7	14.0	18.6	18.3	19.4	23.7	24.0	23.2	21.7	19.2	17.3	12.9	(30)	
	5%	WB	2.4	3.1	5.4	7.6	10.6	14.5	16.0	15.6	13.6	9.0	5.0	2.4	(31)	
		MCDB	10.9	12.4	15.5	17.6	19.3	23.1	23.9	23.3	21.2	18.5	15.0	11.0	(32)	
	10%	WB	1.0	1.7	4.4	6.5	9.9	13.7	15.5	14.9	12.8	8.0	3.8	1.0	(33)	
		MCDB	8.3	9.8	13.9	16.1	18.8	22.9	23.7	22.9	21.2	17.5	12.5	8.2	(34)	
(35) Mean Daily Temperature Range	MDBR		14.0	13.6	14.7	14.2	13.6	15.9	14.6	14.2	14.5	14.7	14.2	13.8	(35)	
	5% DB	MCDBR	16.9	16.8	18.1	18.4	17.7	19.1	17.1	16.4	17.2	18.5	17.3	17.8	(36)	
		MCWBR	8.9	8.3	8.2	7.7	6.3	5.7	4.4	4.7	5.4	7.3	8.0	9.6	(37)	
	5% WB	MCDBR	16.4	16.7	17.1	16.9	15.7	15.2	14.0	13.4	13.5	16.1	16.6	17.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.225	0.230	0.250	0.275	0.293	0.305	0.331	0.321	0.287	0.254	0.234	0.226	(40)	
	taud		2.645	2.660	2.601	2.535	2.546	2.529	2.501	2.520	2.588	2.667	2.642	2.633	(41)	
	Ebn at Noon		994	1027	1025	1005	985	968	940	945	966	980	971	967	(42)	
	Edh at Noon		65	74	88	100	101	103	105	101	88	73	65	61	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.67	3.60	4.86	5.91	6.57	7.34	7.01	6.36	5.44	4.06	2.95	2.37	(44)	
	RadStd		0.14	0.23	0.35	0.37	0.49	0.32	0.29	0.20	0.27	0.24	0.17	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47) Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+70	+33	(47)	

Nomenclature: See separate page